

Automatica Relevance Determination and extensions

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Abstract

Some brief notes on deducing Relevant Vector Machines and Automatic Relevance Determination from the evidence. The novelty is possibly in the low noise limit where we can clearly understand the admission and pruning of basis functions. The sequential algorithm is also described, and the low noise limit is used to show when it will recover the exact basis.

1 Introduction

Given some data $\mathbf{y}$ and a basis $\phi_i(\mathbf{x})$, we consider the linear model

$$y(\mathbf{x}) = \sum_{m=1}^M a_m \phi_m(\mathbf{x}) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2(\mathbf{x})), \quad (1)$$

where $\sigma_\epsilon(\mathbf{x})^2$ is the noise covariance. Evaluating the above for a set of inputs $\mathbf{x}_n$ gives the matrix-vector form

$$\mathbf{y} = \Phi \mathbf{a} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma_\epsilon), \quad (2)$$

where $\mathbf{y} = [y(\mathbf{x}_1), \dots, y(\mathbf{x}_N)]^\top$, Φ is the design matrix with elements $\Phi_{nm} = \phi_m(\mathbf{x}_n)$, $\mathbf{a} = [a_1, \dots, a_M]^\top$, and $\Sigma_\epsilon = \text{diag}(\sigma_\epsilon^2(\mathbf{x}_1), \dots, \sigma_\epsilon^2(\mathbf{x}_N))$. From the above we have the likelihood

$$p(\mathbf{y}|\mathbf{a}) = \mathcal{N}(\mathbf{y}|\Phi \mathbf{a}, \Sigma_\epsilon). \quad (3)$$

If the number of basis functions M is larger than quantity of data N , we want to efficiently determine which basis functions are most relevant. We do this with ARD = Automatic Relevance Determination.

Assume independent Gaussian priors on the coefficients a_i with precision α_i , then by using Bayes theorem followed by completing the square (calculations commented out), the posterior distribution is

$$p(\mathbf{a}|\mathbf{y}) = \mathcal{N}(\mathbf{a}|\boldsymbol{\mu}, \Sigma), \quad (4)$$

where

$$\Sigma = (\Phi^\top \Sigma_\epsilon^{-1} \Phi + \mathbf{A})^{-1}, \quad \boldsymbol{\mu} = \Sigma \Phi^\top \Sigma_\epsilon^{-1} \mathbf{y}, \quad (5)$$

and $\mathbf{A} = \text{diag}(\alpha_1, \alpha_2, \dots)$.

By integrating the joint distribution $p(\mathbf{y}, \mathbf{a}) = p(\mathbf{y}|\mathbf{a})p(\mathbf{a})$ over $\mathbf{a}$ we compute the evidence:

$$p(\mathbf{y}) = \frac{|\Sigma|^{1/2} |\mathbf{A}|^{1/2}}{(2\pi)^{N/2} |\Sigma_\epsilon|^{1/2}} \exp\left(-\frac{1}{2} \mathbf{y}^\top \Sigma_\epsilon^{-1} \mathbf{y} + \frac{1}{2} \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}\right), \quad (6)$$

which is the quantity maximised over the α_m in ARD.

1.1 General solution

When $M > N$ the M columns $\phi_m \in \mathbb{R}^N$ cannot be an independent set of vectors, since at most N vectors can be. Marginalising $\mathbf{a}$ out of $p(\mathbf{y}|\mathbf{a})p(\mathbf{a})$ directly, using that $\mathbf{y} = \Phi \mathbf{a} + \boldsymbol{\epsilon}$ is a sum of independent Gaussians, gives $p(\mathbf{y}) = \mathcal{N}(\mathbf{y}|\mathbf{0}, \mathbf{C})$ with

$$\mathbf{C} = \Sigma_\epsilon + \Phi \mathbf{A}^{-1} \Phi^\top = \beta^{-1} \mathbf{I} + \sum_m \alpha_m^{-1} \phi_m \phi_m^\top, \quad (7)$$

that is

$$p(\mathbf{y}) = \frac{|\mathbf{C}|^{-1/2}}{(2\pi)^{N/2}} \exp\left(-\frac{1}{2} \mathbf{y}^\top \mathbf{C}^{-1} \mathbf{y}\right). \quad (8)$$

This is equivalent to (6). When $M > N$, which does not really happen with the sequential greedy algorithm, it is more useful as $\mathbf{C}$ is $N \times N$ however large M is. One clear advantage is that each basis function enters it through a single rank-one term $\alpha_m^{-1} \phi_m \phi_m^\top$ so that sending $\alpha_m \rightarrow \infty$ deletes that term entirely and prunes ϕ_m from the model. Throughout we take the noise to be isotropic, $\Sigma_\epsilon = \beta^{-1} \mathbf{I}$, and normalise the basis functions so that $\|\phi_m\| = 1$.

To prepare for the sequential algorithm we need consider that Φ contains a subset of all the basis functions. The process below shows how to either update the precision α_m of a basis already in the set, delete it by setting $\alpha_m \rightarrow \infty$, or add a new basis function by introducing a finite α_m .

To update or add α_m , Split off the rank-one contribution of ϕ_m in (7),

$$\mathbf{C} = \mathbf{C}_{-m} + \alpha_m^{-1} \phi_m \phi_m^\top, \quad \mathbf{C}_{-m} := \beta^{-1} \mathbf{I} + \sum_{j \neq m} \alpha_j^{-1} \phi_j \phi_j^\top. \quad (9)$$

For what follows, $\mathbf{C}_{-m}$ could be anything, as long as it is invertible, and the split (9)₁ holds. This will allow us to use ARD for other forms of the evidence covariance.

The matrix determinant lemma and the Sherman–Morrison formula applied to (9) give

$$|\mathbf{C}| = |\mathbf{C}_{-m}| \frac{\alpha_m + s_m}{\alpha_m}, \quad (10)$$

$$\mathbf{C}^{-1} = \mathbf{C}_{-m}^{-1} - \frac{\mathbf{C}_{-m}^{-1} \phi_m \phi_m^\top \mathbf{C}_{-m}^{-1}}{\alpha_m + s_m} \implies \mathbf{y}^\top \mathbf{C}^{-1} \mathbf{y} = \mathbf{y}^\top \mathbf{C}_{-m}^{-1} \mathbf{y} - \frac{q_m^2}{\alpha_m + s_m}, \quad (11)$$

where we defined the *sparsity* s_m and the *quality* q_m of the basis function ϕ_m , **tipping2003fast**

$$s_m := \phi_m^\top \mathbf{C}_{-m}^{-1} \phi_m > 0 \quad \text{and} \quad q_m := \phi_m^\top \mathbf{C}_{-m}^{-1} \mathbf{y}. \quad (12)$$

Substituting (10) and (11) into (8), every term involving α_m collects into a single function of it,

$$\log p(\mathbf{y}) = \mathcal{L}_{-m} + \ell(\alpha_m), \quad \ell(\alpha) = \frac{1}{2} \left[\log \alpha - \log(\alpha + s_m) + \frac{q_m^2}{\alpha + s_m} \right], \quad (13)$$

with $\mathcal{L}_{-m}$ the log-evidence of the model without ϕ_m , which is independent of α_m . Fortunately, we can explicitly calculate the α_m which is the global maximum of $\ell(\alpha_m)$. To do so, we differentiate (13) with respect to α to reach

$$\frac{d\ell}{d\alpha} = \frac{1}{2} \frac{\alpha(s_m - q_m^2) + s_m^2}{\alpha(\alpha + s_m)^2}, \quad (14)$$

whose numerator is a straight line in α , positive at $\alpha = 0$. If $q_m^2 > s_m$ it changes sign once and ℓ has an interior maximum; if $q_m^2 \leq s_m$ it never changes sign, ℓ increases monotonically, and the maximum is at $\alpha_m \rightarrow \infty$. Hence

$$\alpha_m = \frac{s_m^2}{q_m^2 - s_m} \quad \text{if } q_m^2 > s_m, \quad \alpha_m \rightarrow \infty \quad \text{otherwise.} \quad (15)$$

Below we show how this result leads to an efficient sequential algorithm to determine the basis, and in Section 1.3 we give an intuitive explanation of why it works by taking the small noise limit.

1.2 A sequential algorithm

Because each application of (15) maximises the exact evidence over a single α_m with all the others held fixed, sweeping it across candidates is a coordinate ascent on $\log p(\mathbf{y})$: every add, delete or re-estimate increases the evidence, which is bounded for fixed β , so the iteration converges. **tipping2003fast** To choose the best move at each step we need the value of the maximum of (13). Substituting $\alpha_m = s_m^2/(q_m^2 - s_m)$ into ℓ gives

$$\ell_m^* = \frac{1}{2} \left[\frac{q_m^2 - s_m}{s_m} - \log \frac{q_m^2}{s_m} \right] \geq 0, \quad (16)$$

and with the convention $\ell(\infty) = 0$ for a pruned function, the evidence gained by any of the three moves is the single expression

$$\Delta \ell_m = \ell(\alpha_m^{\text{new}}) - \ell(\alpha_m^{\text{old}}), \quad (17)$$

covering addition ($\alpha_m^{\text{old}} = \infty$), deletion ($\alpha_m^{\text{new}} = \infty$) and re-estimation.

The remaining question is cost: taken literally, s_m and q_m require $\mathbf{C}_{-m}^{-1}$, a different $N \times N$ inverse for every candidate. Two identities remove this. First, maintain instead the pair computed from the covariance $\mathbf{C}$ of the *current* model, common to all candidates,

$$S_m := \phi_m^\top \mathbf{C}^{-1} \phi_m, \quad Q_m := \phi_m^\top \mathbf{C}^{-1} \mathbf{y}; \quad (18)$$

applying Sherman–Morrison to $\mathbf{C}_{-m} = \mathbf{C} - \alpha_m^{-1} \phi_m \phi_m^\top$ converts them exactly,

$$s_m = \frac{\alpha_m S_m}{\alpha_m - S_m}, \quad q_m = \frac{\alpha_m Q_m}{\alpha_m - S_m}, \quad (19)$$

which reduces to $s_m = S_m, q_m = Q_m$ for a pruned ϕ_m . Second, when only the functions in a set S are retained, $|S| = k$, the Woodbury identity applied to (7) gives

$$\mathbf{C}^{-1} = \beta \mathbf{I} - \beta^2 \Phi_S \Sigma_S \Phi_S^\top, \quad (20)$$

where $\Sigma_S = (\mathbf{A}_S + \beta \Phi_S^\top \Phi_S)^{-1}$ and $\mu_S = \beta \Sigma_S \Phi_S^\top \mathbf{y}$ are the posterior covariance and mean (5) restricted to the retained set, so that

$$S_m = \beta - \beta^2 (\Phi_S^\top \phi_m)^\top \Sigma_S (\Phi_S^\top \phi_m), \quad Q_m = \beta \phi_m^\top (\mathbf{y} - \Phi_S \mu_S). \quad (21)$$

Everything is assembled from the $k \times k$ matrix Σ_S and the current residual $\mathbf{y} - \Phi_S \mu_S$; no $N \times N$ or $M \times M$ matrix is ever formed. The second formula also exposes what the algorithm is doing: Q_m is β times the correlation of ϕ_m with the *residual*, so the sweep is a matching pursuit on residuals, with S_m discounting whatever the retained functions already explain. Given the inner products $\phi_j^\top \phi_m$ and $\phi_m^\top \mathbf{y}$, one sweep over all M candidates costs $O(Mk^2)$, and after each accepted move Σ_S, μ_S and every pair (S_m, Q_m) can be refreshed by rank-one updates rather than recomputed. **tipping2003fast** Algorithm 1 collects the loop. If the noise level is unknown, β can be re-estimated between sweeps from $\beta^{-1} = \|\mathbf{y} - \Phi_S \mu_S\|^2 / (N - k + \sum_{j \in S} \alpha_j [\Sigma_S]_{jj})$.

Algorithm 1 Sequential ARD by greedy evidence maximisation. **tipping2003fast**

Require: data $\mathbf{y}$, candidates $\{\phi_m\}_{m=1}^M$ with $\|\phi_m\| = 1$, noise precision β

- 1: $m_1 \leftarrow \arg \max_m (\phi_m^\top \mathbf{y})^2$; $S \leftarrow \{m_1\}$; $\alpha_{m_1}^{-1} \leftarrow (\phi_{m_1}^\top \mathbf{y})^2 - \beta^{-1}$
 - 2: compute Σ_S, μ_S and (S_m, Q_m) for all m from (21)
 - 3: **repeat**
 - 4: $(s_m, q_m) \leftarrow (19)$ for all m
 - 5: $\alpha_m^{\text{new}} \leftarrow (15)$ and $\Delta \ell_m \leftarrow (17)$ for all m
 - 6: $m^* \leftarrow \arg \max_m \Delta \ell_m$
 - 7: (Optional) use gradient descent to adjust ϕ_{m^*}
 - 8: apply the move for m^* : add it to S , delete it from S , or re-estimate α_{m^*}
 - 9: refresh Σ_S, μ_S and all (S_m, Q_m) by rank-one updates
 - 10: **until** $\Delta \ell_{m^*}$ falls below tolerance
 - 11: **return** retained set S , coefficients μ_S , covariance Σ_S , precisions α_j
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1.3 Low noise limit

Every move of the sequential algorithm — add, delete or re-estimate — is decided by the sparsity s_m and the quality q_m of (12). Here we perform an asymptotic for low noise $\beta^{-1} \rightarrow 0$ to rewrite s_m and q_m as an intuitive representation. We then specialise the data $\mathbf{y}$ to be composed exactly of a subset of the basis, and use the same limit to determine when ARD recovers that subset exactly.

Consider a model that retains the set S , whose columns Φ_S we take to be linearly independent, with rank $r = |S|$, so that its covariance (7) is

$$\mathbf{C}_S = \beta^{-1} \mathbf{I} + \Phi_S \mathbf{A}_S^{-1} \Phi_S^\top. \quad (22)$$

The rank- r matrix $\Phi_S \mathbf{A}_S^{-1} \Phi_S^\top$ acts only within $\text{span}(\Phi_S)$, so $\mathbf{C}_S$ is block diagonal with respect to the split of $\mathbb{R}^N$ into $\text{span}(\Phi_S)$, where its eigenvalues are $O(1)$, and the $(N - r)$ -dimensional orthogonal complement where every eigenvalue equals β^{-1} . Inverting each block separately gives

$$\mathbf{C}_S^{-1} = \beta \Phi_S^\perp (\Phi_S^\perp)^\top + \mathbf{K}^+ + O(\beta^{-1}), \quad \text{with} \quad \mathbf{K} = \Phi_S \mathbf{A}_S^{-1} \Phi_S^\top, \quad (23)$$

where

$$\mathbf{K}^+ = \Phi_S \mathbf{G}^{-1} \mathbf{A}_S \mathbf{G}^{-1} \Phi_S^\top, \quad \text{with} \quad \mathbf{G} = \Phi_S^\top \Phi_S.$$

We can test the expansion (23) is correct by multiplying (23) by (22) and then using $\Phi_S^\perp (\Phi_S^\perp)^\top + \Phi_S (\Phi_S)^\top = \mathbf{I}$.

For a new candidate ϕ_m with $m \notin S$ we set $\mathbf{C}_{-m} = \mathbf{C}_S$, and substituting (23) into the definitions (12) gives

$$s_m = \beta \|\phi_m^\perp\|^2 + \phi_m^\top \mathbf{K}^+ \phi_m + O(\beta^{-1}), \quad q_m = \beta (\phi_m^\perp)^\top \mathbf{y}^\perp + \phi_m^\top \mathbf{K}^+ \mathbf{y} + O(\beta^{-1}), \quad (24)$$

where

$$\phi_m^\perp := \Phi_S^\perp (\Phi_S^\perp)^\top \phi_m \quad \text{and} \quad \mathbf{y}^\perp := \Phi_S^\perp (\Phi_S^\perp)^\top \mathbf{y},$$

where we used that $(\phi_m^\perp)^\top \mathbf{y} = (\phi_m^\perp)^\top \mathbf{y}^\perp$. The term $\phi_m^\perp$ is the *novel component* of the candidate and $\mathbf{y}^\perp$ is the *residual* — the part of the data the basis Φ_S cannot produce.

Novel component

For a novel component $\|\phi_m^\perp\| > 0$ the leading order term has a simple interpretation: the sparsity s_m measures how much of ϕ_m is genuinely new, while the quality q_m measures how strongly that new part correlates with the unexplained data. The admission test $q_m^2 > s_m$ of (15) becomes, at leading order,

$$(\hat{\phi}_m^\perp{}^\top \mathbf{y})^2 > \beta^{-1}, \quad \hat{\phi}_m^\perp := \phi_m^\perp / \|\phi_m^\perp\| : \quad (25)$$

project the data onto the unit vector along the candidate's novel direction, and admit the candidate only if that component exceeds one noise standard deviation $\beta^{-1/2}$. In this limit ARD is a matching pursuit on residuals with a noise-scaled admission threshold. When the test is passed, the optimal precision (15) simplifies to

$$\alpha_m^{-1} = \frac{q_m^2 - s_m}{s_m^2} \rightarrow \left(\frac{(\phi_m^\perp)^\top \mathbf{y}}{\|\phi_m^\perp\|^2} \right)^2, \quad (26)$$

and the ratio on the right is the same as the least-squares coefficient obtained by fitting the residual with the single regressor $\phi_m^\perp$. ARD grants each admitted function a prior variance equal to the square of the coefficient it is about to take — just enough room to hold it, and no more.

A basis in the span

When a new candidate ϕ_m is in $\text{span}(\Phi_S)$ then $\phi_m^\perp = \mathbf{0}$ and the leading order terms in (24) are zero, and so we must use the next order terms to check $q_m^2 > s_m$. In the next order the orthogonal projection $\mathbf{y}^\perp$ is ignored (because $\mathbf{K}^+ \mathbf{y}^\perp = \mathbf{0}$), so it helps to use the representation

$$\mathbf{y} = \Phi_S \mathbf{c} + \mathbf{y}_\perp \quad \text{and} \quad \phi_m = \Phi_S \mathbf{b}. \quad (27)$$

Using this representation, we assume all previously retained functions had their α_j set by ARD from (26) given by $\alpha_j^{-1} = c_j^2$ which is calculated by using

$$(\phi_j^\perp)^\top \mathbf{y} = (\phi_j^\perp)^\top \Phi_S \mathbf{c} = \|\phi_j^\perp\|^2 c_j,$$

where just for the equation above in this section we have that $\phi_j^\perp := \Phi_{S \setminus j}^\perp (\Phi_{S \setminus j}^\perp)^\top \phi_j$ for all $j \in S$. With these α_j the posterior mean is $\boldsymbol{\mu}_S = (\Phi_S^\top \Phi_S + \beta^{-1} \mathbf{A}_S)^{-1} \Phi_S^\top \Phi_S \mathbf{c} = \mathbf{c} + O(\beta^{-1})$, the correct coefficients up to a shrinkage of order β^{-1} .

Substituting the above into (24) gives the next order terms

$$q_m = \mathbf{b}^\top \Phi_S^\top \mathbf{K}^+ \mathbf{y} = \mathbf{b}^\top \mathbf{A}_S \mathbf{c}, \quad s_m = \mathbf{b}^\top \Phi_S^\top \mathbf{K}^+ \Phi_S \mathbf{b} = \mathbf{b}^\top \mathbf{A}_S \mathbf{b}, \quad (28)$$

and the admission test $q_m^2 > s_m$ becomes

$$(\mathbf{b}^\top \mathbf{A}_S \mathbf{c})^2 > \mathbf{b}^\top \mathbf{A}_S \mathbf{b}. \quad (29)$$

Now we substitute $\alpha_j = c_j^{-2}$ into $\mathbf{A}_S$ and open up both sides into a sum to reach

$$\left(\sum_{j \in S} u_j \right)^2 > \sum_{j \in S} u_j^2, \quad \text{with} \quad u_j := \frac{b_j}{c_j}, \quad (30)$$

where $\phi_m = \sum_{j \in S} u_j (c_j \phi_j)$, so that u_j is the fraction of the retained contribution $c_j \phi_j$ that the candidate carries, and $\mathbf{u} \propto \mathbf{1}$ precisely when $\phi_m \parallel \mathbf{y}$. Both sides are quadratic in $\mathbf{u}$, so only its direction matters: writing θ for the angle between $\mathbf{u}$ and $\mathbf{1}$, so that $(\cos \theta)^2 = (\sum_j u_j)^2 / (k \sum_j u_j^2)$, and the test (30) becomes $k \cos^2 \theta > 1$. Equivalently, define the *effective number of retained functions the candidate merges*,

$$k_m^{\text{eff}} := \frac{q_m^2}{s_m} = \frac{(\sum_j u_j)^2}{\sum_j u_j^2} \leq k, \quad (31)$$

by Cauchy–Schwarz, with equality when every u_j is equal. The admission test is then simply $k_m^{\text{eff}} > 1$: a candidate already in the span earns its place only by compressing more than one function's worth. Its precision (15) and evidence gain (16) follow at once,

$$\alpha_m^{-1} = \frac{k_m^{\text{eff}} - 1}{s_m}, \quad \ell_m^* = \frac{1}{2} [k_m^{\text{eff}} - 1 - \log k_m^{\text{eff}}], \quad (32)$$

independent of the inner products among the retained functions and of the normalisation of ϕ_m . Note this requires $c_j \neq 0$ for every $j \in S$; a candidate with $b_j \neq 0$ where $c_j = 0$ has $s_m \rightarrow \infty$ and is rejected outright.

Shedding the originals. If the retained set now has $k + 1$ members while still spanning only k dimensions, and since $\mathbf{y} = \|\mathbf{y}\| \phi_d$, every one of the original ϕ_j can be deleted without making $\mathbf{y}$ unreachable. By (??) the evidence at leading order counts the dimensions spanned, so each deletion that lowers the rank is worth $\frac{1}{2} \log \beta$, and reducing the retained set from S^* to $\{d\}$ lowers the rank from k to 1:

$$\log p(\mathbf{y} \mid \{d\}) - \log p(\mathbf{y} \mid S^*) = \frac{k-1}{2} \log \beta + O(1). \quad (33)$$

The greedy sweep therefore has a strictly uphill path from the fat interpolating model to the sparse one — a single $O(1)$ add, followed by deletions worth $\frac{1}{2} \log \beta$ apiece — and once the originals are gone the decoy relaxes to $\alpha_d^{-1} = \|\mathbf{y}\|^2$, its own coefficient squared.

Why this is not a guarantee. The evidence always prefers the smaller basis, but Algorithm 1 is coordinate ascent: it reaches that basis only when a chain of *individually* uphill single moves leads there. The decoy above is reachable because one function alone replaces all k originals. If instead the sparser description needs two new functions to displace three retained ones, and neither new function passes (29) on its own, the sweep stalls at a local maximum and the shorter description is never found. The per-candidate test sees one swap at a time, and the evidence landscape it climbs is not concave. **wipf2008new**